

Strategic Behaviour of AI Agents in Noisy Multi-Player Social Delimmas



An empirical analysis of how cooperation, volatility and reward shifts across scaling, action granularity, noise and streak incentives



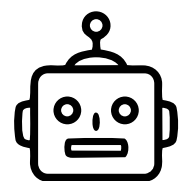
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1 Motivation and Research Question

How does strategic behavior of LLM agents transform as we move from deterministic settings to noisy, stochastic, multi-player social dilemmas?

- We analyze cooperation, reward, confidence, and volatility across four experimental axes: number of agents (N), action granularity (M), perception noise (p), and streak threshold (θ).
- The goal is to identify when LLM policies remain stable and when richer environments induce divergence.

2 Experimental Setup



Agents: GPT-OSS 20B, Llama 3.1 8B, DeepSeek v3.2, Gemma 3 27B, Qwen3 235B



Parameters analyzed: N, M, p, θ



Core metrics: mean cooperation, average reward, confidence, action volatility

Logs aggregated across multiple experimental runs for model-wise comparison

3 Model Summary Table

Table 1. Aggregate performance summary across all experimental conditions.

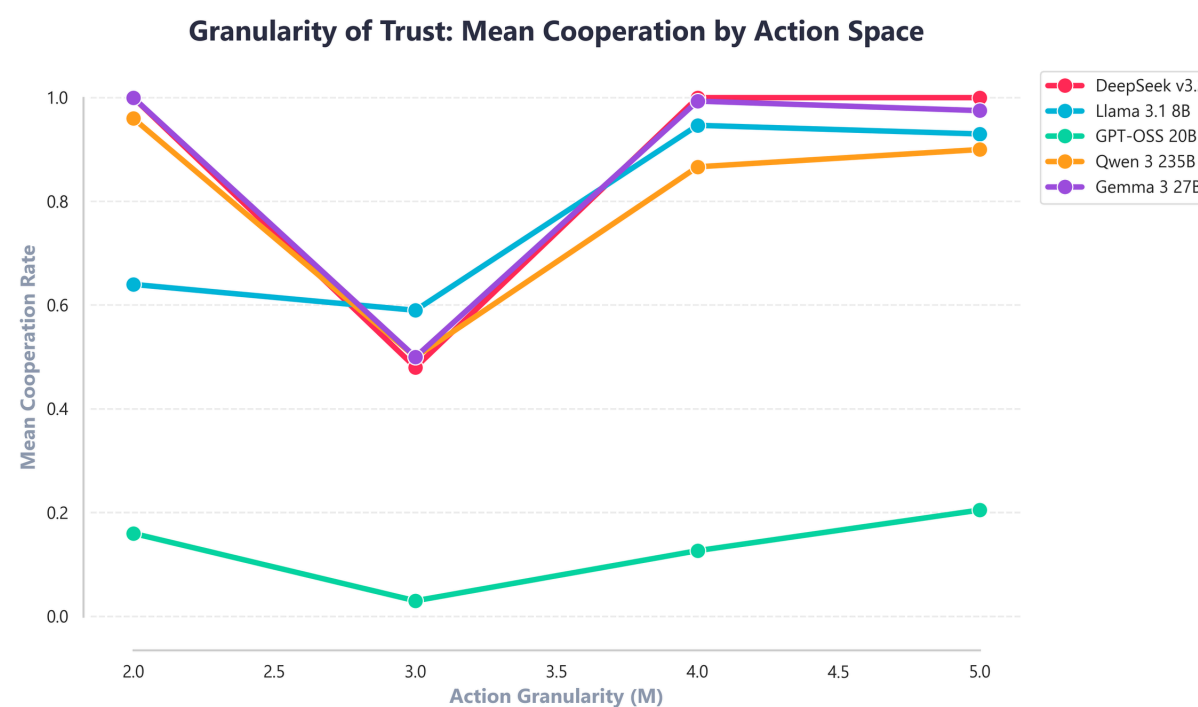
Model	Coop Rate	Avg Reward	Confidence	Volatility
Llama 3.1 8B	83.90%	1.61	0.791	33.60%
Gemma 3 27B	82.40%	2.68	0.836	8.40%
DeepSeek v3.2	81.10%	2.4	0.823	14.00%
Qwen3 235B	77.20%	3.55	0.877	15.00%
GPT-OSS 20B	6.30%	11.18	0.846	13.20%

5 Insights & Takeaways

- Aggregate cooperation alone can hide distinct strategic styles.
- Llama 3.1 8B is the most consistently cooperative model.
- Qwen3 235B offers the strongest balance between reward and cooperation among cooperative models.
- GPT-OSS 20B is the major outlier: very low cooperation but highest reward, indicating payoff-maximizing defection.

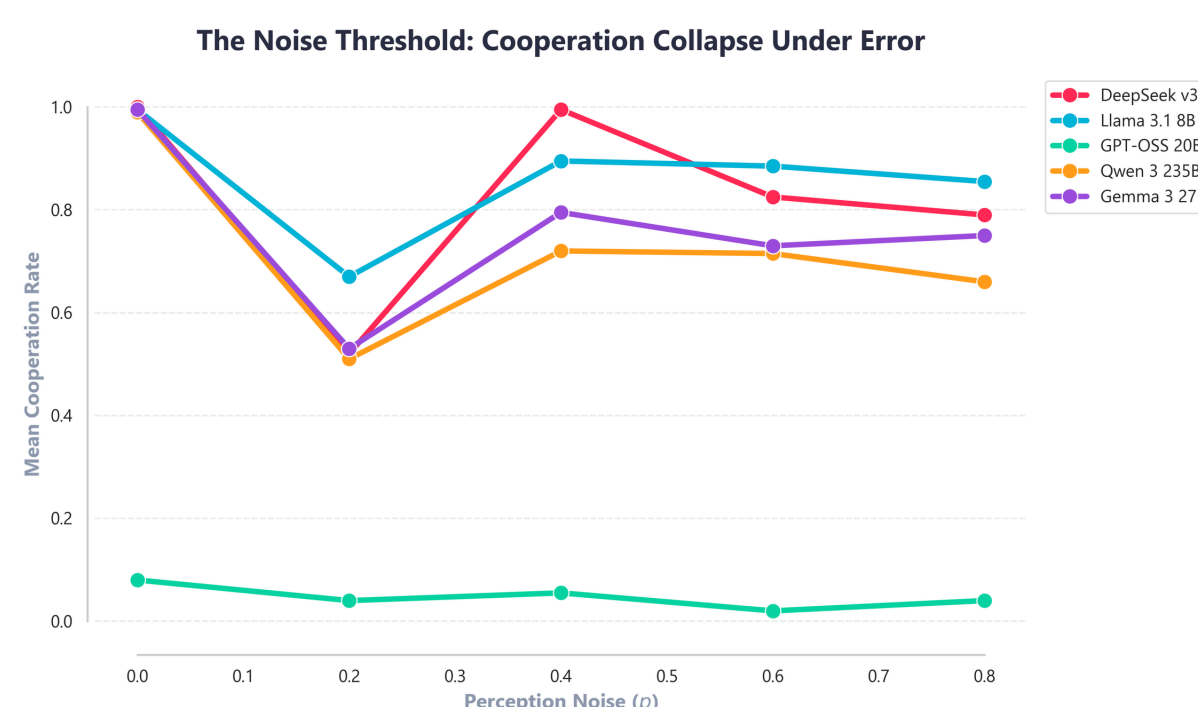
4 Key Results

i Granularity of Trust: Mean Cooperation by Action Space



Cooperation dips at M=3 for most models, then rebounds sharply as the action space expands, suggesting finer control can restore trust.

ii The Noise Threshold: Cooperation Collapse Under Error



Even small observation noise sharply reduces cooperation for most models before partial recovery at higher noise levels. GPT-OSS remains near-defective throughout.

v Strategy Volatility: Probability of Changing Action

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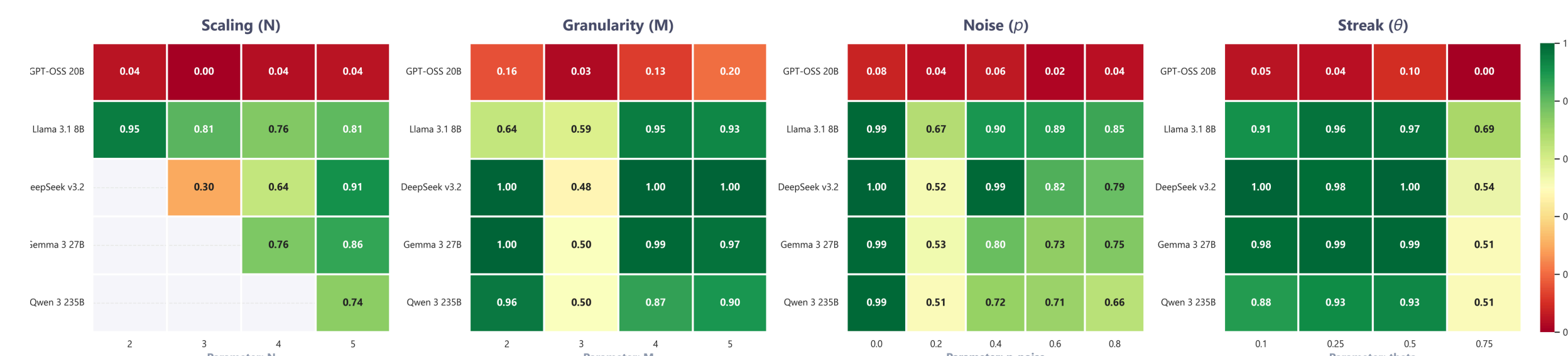


Volatility is highly model- and parameter-dependent: Llama is the most reactive across many settings, while Gemma is comparatively stable.

The heatmap summarizes model-wise cooperation patterns across all parameter axes, highlighting broad stability for Llama, DeepSeek, Gemma, and Qwen, versus persistent non-cooperation from GPT-OSS.

vi Behavioral Profiling: Mean Cooperation Rate by LLM and Parameter

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Conclusion

- While large LLM swarms naturally collapse into the strict Nash Equilibrium of total defection, expanding action granularity allows them to safely bypass this mathematical trap and organically discover cooperative, Pareto-optimal states.
- Parameter size heavily influences gullibility. Smaller models (like Llama 8B) default to naive, prosocial cooperation and are easily ruthlessly exploited by larger, more cynical models (like Qwen 235B)

